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Devices and methods for leaflet cutting

Abstract

A system for cutting leaflet tissue at a cardiac valve may comprise a guide catheter having a proximal end and a distal end, wherein the distal end of the guide catheter is steerable to a position above a cardiac valve. The system may also include a handle coupled to the proximal end of the guide catheter, the handle comprising at least one control configured to steer the guide catheter to the position above the cardiac valve. Finally, the system may comprise a cutting mechanism routable through the guide catheter and able to be positioned at the distal end of the guide catheter, the cutting mechanism configured to cut a portion of leaflet tissue of the cardiac valve.

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6718985	12/2003	Hlavka et al.	N/A	N/A
6719767	12/2003	Kimblad	N/A	N/A
6723038	12/2003	Schroeder et al.	N/A	N/A
6726716	12/2003	Marquez	N/A	N/A
6740107	12/2003	Loeb et al.	N/A	N/A
6746471	12/2003	Mortier et al.	N/A	N/A
6752813	12/2003	Goldfarb et al.	N/A	N/A
6755777	12/2003	Schweich et al.	N/A	N/A
6764510	12/2003	Vidlund et al.	N/A	N/A
6767349	12/2003	Ouchi	N/A	N/A
6770083	12/2003	Seguin	N/A	N/A
6797001	12/2003	Mathis et al.	N/A	N/A
6797002	12/2003	Spence et al.	N/A	N/A
6860179	12/2004	Hopper et al.	N/A	N/A
6875224	12/2004	Grimes	N/A	N/A
6926715	12/2004	Hauck et al.	N/A	N/A
6932810	12/2004	Ryan	N/A	N/A
6945978	12/2004	Hyde	N/A	N/A
6949122	12/2004	Adams et al.	N/A	N/A
6966914	12/2004	Abe	N/A	N/A
6986775	12/2005	Morales et al.	N/A	N/A
7004970	12/2005	Cauthen et al.	N/A	N/A
7011669	12/2005	Kimblad	N/A	N/A
7033390	12/2005	Johnson et al.	N/A	N/A
7048754	12/2005	Martin et al.	N/A	N/A
7056294	12/2005	Khairkhahan	606/167	A61B 17/32075
7112207	12/2005	Allen et al.	N/A	N/A
7226467	12/2006	Lucatero et al.	N/A	N/A
7258694	12/2006	Choi et al.	N/A	N/A
7288097	12/2006	Seguin	N/A	N/A
7291168	12/2006	Macoviak et al.	N/A	N/A
7338467	12/2007	Lutter	N/A	N/A
7381210	12/2007	Zarbatany et al.	N/A	N/A
7435257	12/2007	Lashinski et al.	N/A	N/A
7464712	12/2007	Oz et al.	N/A	N/A
7497822	12/2008	Kugler et al.	N/A	N/A
7533790	12/2008	Knodel et al.	N/A	N/A
7563267	12/2008	Goldfarb et al.	N/A	N/A
7563273	12/2008	Goldfarb et al.	N/A	N/A
7604646	12/2008	Goldfarb et al.	N/A	N/A
7608091	12/2008	Goldfarb et al.	N/A	N/A
7635329	12/2008	Goldfarb et al.	N/A	N/A

7651502	12/2009	Jackson	N/A	N/A
7655015	12/2009	Goldfarb et al.	N/A	N/A
7666204	12/2009	Thornton et al.	N/A	N/A
7955340	12/2010	Michlitsch et al.	N/A	N/A
8216234	12/2011	Long	N/A	N/A
8257356	12/2011	Bleich et al.	N/A	N/A
8398708	12/2012	Meiri et al.	N/A	N/A
8435237	12/2012	Bahney	N/A	N/A
8496655	12/2012	Epp et al.	N/A	N/A
8500768	12/2012	Cohen	N/A	N/A
8523881	12/2012	Cabiri et al.	N/A	N/A
8623077	12/2013	Cohn	N/A	N/A
8690858	12/2013	Machold et al.	N/A	N/A
8821518	12/2013	Saliman et al.	N/A	N/A
8926588	12/2014	Berthiaume et al.	N/A	N/A
9126032	12/2014	Khairkhanan et al.	N/A	N/A
9211119	12/2014	Hendricksen et al.	N/A	N/A
9370341	12/2015	Ceniccola et al.	N/A	N/A
9498331	12/2015	Chang et al.	N/A	N/A
9572666	12/2016	Basude et al.	N/A	N/A
9770256	12/2016	Cohen et al.	N/A	N/A
9949833	12/2017	McCleary et al.	N/A	N/A
10238493	12/2018	Metchik	N/A	A61F 2/2466
10667804	12/2019	Basude et al.	N/A	N/A
11013554	12/2020	Coates	N/A	N/A
11406250	12/2021	Saadat et al.	N/A	N/A
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2002/0161378	12/2001	Downing	N/A	N/A
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2006/0020334	12/2005	Lashinski	623/2.11	A61F 2/014
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/020,669, filed May 6, 2020, the entire contents of which are incorporated by reference herein.

BACKGROUND OF THE INVENTION

(1) The mitral valve controls blood flow from the left atrium to the left ventricle of the heart, preventing blood from flowing backwards from the left ventricle into the left atrium so that it is instead forced through the aortic valve for delivery of oxygenated blood throughout the body. A properly functioning mitral valve opens and closes to enable blood flow in one direction. However, in some circumstances the mitral valve is unable to close properly, allowing blood to regurgitate back into the atrium.

(2) Mitral valve regurgitation has several causes. Functional mitral valve regurgitation is characterized by structurally normal mitral valve leaflets that are nevertheless unable to properly coapt with one another to close properly due to other structural deformations of surrounding heart structures. Other causes of mitral valve regurgitation are related to defects of the mitral valve leaflets, mitral valve annulus, or other mitral valve tissues.

(3) The most common treatments for mitral valve regurgitation rely on valve replacement or repair including leaflet and annulus remodeling, the latter generally referred to as valve annuloplasty. One technique for mitral valve repair which relies on suturing adjacent segments of the opposed valve leaflets together is referred to as the "bowtie" or "edge-to-edge" technique. While these techniques can be effective, they usually rely on open heart surgery where the patient's chest is opened, typically via a sternotomy, and the patient is placed on cardiopulmonary bypass. The need to both, open the chest, and place the patient on bypass, is traumatic and has an associated high mortality and morbidity rate. In some patients, a fixation device can be installed into the heart using minimally invasive techniques. The fixation device can hold the adjacent segments of the opposed valve leaflets together and may reduce mitral valve regurgitation. One such device used to clip the anterior and posterior leaflets of the mitral valve together is the MitraClip® fixation device, sold by Abbott Vascular, Santa Clara, Calif., USA.

(4) However, sometimes after a fixation device is installed, undesirable mitral valve regurgitation can still exist, or can arise again. For patients requiring re-intervention, the presence of a fixation

device in their mitral valves obstructs transcatheter mitral valve replacement. These patients may also be considered too frail to tolerate open-heart surgery, so they are left with no viable options to further improve the function of their mitral valve.

(5) Accordingly, it would be desirable to provide alternative and additional methods, devices, and systems for removing or disabling fixation devices that are already installed in preparation for the installation of an artificial, replacement mitral valve. The methods, devices, and systems may be useful for repair of tissues in the body other than heart valves. At least some of these objectives will be met by the invention described hereinbelow.

BRIEF SUMMARY OF THE INVENTION

(6) Implementations of the present invention solve one or more problems in the art with systems, methods, and apparatus configured to cut leaflet tissue at a cardiac valve. The system may comprise a guide catheter having a proximal end and a distal end, wherein the distal end of the guide catheter is steerable to a position above a cardiac valve. The system may also include a handle coupled to the proximal end of the guide catheter, the handle comprising at least one control configured to steer the guide catheter to the position above the cardiac valve. Finally, the system may comprise a cutting mechanism, routable through the guide catheter and able to be positioned at the distal end of the guide catheter, configured to cut a portion of leaflet tissue of the cardiac valve.

(7) A method for cutting leaflet tissue at a cardiac valve within a body may comprise positioning a guide catheter, having a proximal and a distal end such that the distal end of the guide catheter is positioned at a cardiac valve. The method may further comprise routing a cutting mechanism through the guide catheter such that the cutting mechanism extends to the distal end of the guide catheter, wherein the cardiac valve is associated with an interventional implant that approximates adjacent leaflets of the cardiac valve, and a cutting mechanism extends from the guide catheter. Also, the method may include positioning the hook catheter to place the cutting mechanism into contact with leaflet tissue located adjacent to the interventional implant and actuating the cutting mechanism to cut at a portion of least one leaflet of the approximated adjacent leaflet.

(8) Additional features and advantages of exemplary implementations of the invention will be set forth in the description which follows, and in part will be obvious from the description, or may be learned by the practice of such exemplary implementations. The features and advantages of such implementations may be realized and obtained by means of the instruments and combinations particularly pointed out in the appended claims. These and other features will become more fully apparent from the following description and appended claims or may be learned by the practice of such exemplary implementations as set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to describe the manner in which the above-recited and other advantages and features of the invention can be obtained, a more particular description of the invention briefly described above will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. Understanding that these drawings depict only typical embodiments of the invention and are not therefore to be considered to be limiting of its scope, the invention will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

(2) FIG. 1 illustrates an embodiment of a delivery system that may be utilized for guiding and/or delivering a cutting mechanism to a cardiac valve;

(3) FIG. 2 is a schematic representation of a human heart with an interventional implant affixed between the anterior and posterior leaflets of the mitral valve and a guiding catheter extending from a transseptal puncture and positioned above the mitral valve;

- (4) FIGS. 3A-3D illustrate an embodiment of a cutting mechanism according to the present disclosure shown in use in association with a cardiac valve;
- (5) FIGS. 4A-4F illustrate an alternative embodiment of a cutting mechanism according to the present disclosure;
- (6) FIGS. 5A-5J illustrate an alternative embodiment of a cutting mechanism according to the present disclosure shown in use in association with a cardiac valve;
- (7) FIGS. 6A-6F illustrate an alternative embodiment of a cutting mechanism according to the present disclosure shown in use in association with a cardiac valve;
- (8) FIGS. 7A-7D illustrate alternative embodiment of a cutting mechanism according to the present disclosure shown in use in association with a cardiac valve; and
- (9) FIGS. 8A-8C illustrate yet another embodiment of a cutting mechanism according to the present disclosure shown in use in association with a cardiac valve.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(10) Implementations of the present invention solve one or more problems in the art with systems, methods, and apparatus configured to cut leaflet tissue at a cardiac valve. More specifically, at least one embodiment of the present invention, may comprise a guide catheter having a proximal end and a distal end, wherein the distal end of the guide catheter is steerable to a position above a cardiac valve. The system may also include a handle coupled to the proximal end of the guide catheter, the handle comprising at least one control configured to steer the guide catheter to the position above the cardiac valve. Finally, the system may comprise a cutting mechanism routable through the guide catheter and able to be positioned at the distal end of the guide catheter, the cutting mechanism configured to cut a portion of leaflet tissue of the cardiac valve.

(11) FIG. 1 illustrates an embodiment of a delivery system **100** that may be utilized for guiding and/or delivering a cutting mechanism to the cardiac valve. In at least one embodiment, the delivery system **100** includes a leaflet cutting system **100**, which can include a delivery system **102** that may be utilized for guiding and/or delivering a cutting mechanism to the cardiac valve. The delivery system **102** can include a guide catheter **105** having a proximal end **112** and a distal end **115**. The delivery system may comprise a handle **110** positioned on the proximal end **112** of the guide catheter **105**. The guide catheter **105** may be operatively coupled to the handle **110**. The guide catheter **105** may include a steerable portion **117** near the distal end **115** that can be steerable to enable the guiding and orienting of the guide catheter **105** through a patient's vasculature to a targeted treatment site, such as a mitral valve. For example, the handle **110** may include at least one control **120** (e.g., a dial, a switch, a slider, a button, etc.) that can be actuated to control the movement and curvature of a steerable portion **117** of the guide catheter **105**.

(12) In at least one embodiment, the at least one control **120** can be operatively coupled to one or more control lines **125** (e.g., pull wires) extending from the handle **110** through the guide catheter **105** to the distal end **115** of the guide catheter (e.g., through one or more lumens in the guide catheter **105**). Actuation of the at least one control **120** may adjust the tensioning of one or more control lines **125** to steer the guide catheter **105** in a desired curvature and/or direction. FIG. 1 shows the handle **110** as having a single control **120** for providing steerability. Alternatively, a handle **110** may comprise more than one control **120** associated with any number of control lines.

(13) While control lines or wires are described at various points in this application, it should be understood that references made throughout this application to control lines or wires may be a single wire or plurality of wires including or made of steel, titanium alloy, aluminum alloy, nickel alloy, other metals, a shape memory material (such as a shape memory alloy or shape memory polymer), inorganic polymer, organic polymer, ceramic, carbon materials, or other flexible material with sufficient tensile strength. For example, a control line **125** may be a steel cable. In another example, a control line **125** may be a monofilament suture. In another example, a control line **125** may be a multifilament suture. In yet another example, a control line **125** may be a braided suture.

(14) It is desirable for guide catheter **105** to provide an adjustable distal end **115**, which is capable

of being positioned within a target body cavity in a desired orientation. Guide catheter **105** should have a large lumen diameter to accommodate the passage of a variety of devices, such as the various embodiments of the cutting mechanisms discussed hereinafter, and should have good wall strength to avoid kinking or collapse when bent around tight curves, and should have good column, tensile, and torsional strength to avoid deformation when the devices are passed through the lumen and torqued or translated. Guide catheter **105** should provide for a high degree of controlled deflection at its distal end **115**, but should not take up significant lumen area to allow for passage of interventional devices, such as the cutting mechanisms discussed below. Further, guide catheter **105** should be able to be positioned in a manner which allows compound curves to be formed, for example curvature within more than one plane. Such manipulation should also allow fine control over distal end **115** to accommodate anatomical variations within the same type of body cavity and for use in different types of body cavities.

(15) The guide catheter **105** may comprise a main body made of or including a flexible material. The main body may be made of or include a variety of flexible materials, such as thermoplastic elastomers (TPE). In some embodiments, the main body may be a polyether block amide (PEBA or PEBAX). The main body may have a constant durometer or may have varying durometer that varies along its longitudinal length or that varies in different portions of the body. For example, the main body of guide catheter **105** may be made of or include a body material having a durometer of 25D to 75D. In another example, the main body of guide catheter **105** may be made of or include a body material that has a durometer of about 45D. In at least one embodiment, the body material may include PEBAX **4533**. In at least another embodiment, the body material may include PEBAX **3533**.

(16) The guide catheter **105** preferably defines a central lumen, extending axially through its entire length, through which other elongate elements, such as the cutting mechanisms may be inserted for accessing a treatment site. The central lumen may also include a central lumen lining on an inner surface thereof. In some embodiments, the central lumen lining may be a protective material that protects the interior walls from damage due to another element of the elongated member moving through or within the central lumen. In other embodiments, the central lumen lining may include a lubricious coating that reduces friction between the interior wall and another element of the elongated member moving through or within the central lumen. The central lumen lining may include PEB A, polytetrafluoroethylene ("PTFE"), polyetheretherketone ("PEEK"), other polymers, thermoplastic polyurethane ("TPU"), polyethylene with pebble stone surface, silicone oil stainless steel, Nitinol, other metals, or combinations thereof. In at least one embodiment, the central lumen lining may include a plurality of PEBA materials having different durometers.

(17) In other embodiments, the guide catheter **105** may also have an outer layer. In some embodiments, the outer layer may be made of or include a single material or may be made of or include different materials to impart different handling characteristics to the guide catheter **105**. For example, the outer layer may be made of or include softer materials to promote flexibility of the guide catheter **105**. In other examples, the outer layer may be made of or include stiffer materials to promote pushability and/or torqueability of the guide catheter **105**. In yet other examples, the outer layer may include lubricious materials to reduce friction between the guide catheter **105** and the body lumen of the patient. The outer layer may include PEBA, polytetrafluoroethylene ("PTFE"), polyetheretherketone ("PEEK"), other polymers, thermoplastic polyurethane ("TPU"), polyethylene with pebble stone surface, silicone oil stainless steel, Nitinol, other metals, or combinations thereof. In at least one embodiment, the outer layer may include a plurality of PEB A materials having different durometers.

(18) In some embodiments, the outer layer of guide catheter **105** may also include a radiopaque marker to improve visualization of guide catheter **105** during a medical procedure. For example, the outer layer may include a barium sulfate (BaSO₄), gold, platinum, platinum iridium, iodine, other radiopaque materials, or combinations thereof on a distal portion of guide catheter **105**. In at

least one embodiment, one or more additional radiopaque markers may be longitudinally located at one or more intermediate locations along the length of guide catheter **105**.

(19) The curves of guide catheter **105** may be formed by any suitable means. In some embodiments, one or more of the curves are preset so that the curve is formed by shape memory. For example, guide catheter **105** may be comprised of a flexible polymer material in which a curve is preset by heating. When guide catheter **105** is loaded on a guidewire, dilator, obturator or introductory device, the flexibility of guide catheter **105** can allow it to follow the shape or path of the introductory device for proper positioning within the body. When the introductory device is pulled back and/or removed, guide catheter **105** can then resume the shape memory configuration which was preset into the catheter.

(20) Alternatively, the curves may be formed or enhanced with the use of one or more steering mechanisms. In some embodiments, the steering mechanism comprises at least one control wire or pull wire attached to one of the guide catheter **105**, wherein actuation of the steering mechanism applies tension to the at least one pull wire whereby the curve is formed. The pull wires can extend through the central lumen or through individual lumens in the wall of guide catheter **105**. It may be appreciated that more than one pull wire may extend through any given lumen. The presence of each pull wire allows curvature of guide catheter **105** in the direction of the pull wire. For example, when pulling or applying tension to a pull wire extending along one side of the catheter, the catheter will bend, arc or form a curvature toward that side. To then straighten the catheter, the tension may be relieved for recoiling effects or tension may be applied to a pull wire extending along the opposite side of the catheter. Therefore, pull wires are often symmetrically placed along the sides of the catheter.

(21) Thus, in some embodiments at least two pull wires are attached in diametrically opposed locations wherein applying tension to one of the pull wires curves the catheter in one direction and applying tension to the pull wire attached in the diametrically opposed location curves the catheter in another direction opposite to the one direction. The diametrically opposed pull wires may be considered a set. Any number of sets may be present in a catheter to provide unlimited directions of curvature. In some embodiments, the steering mechanism can comprise at least four pull wires wherein two of the at least four pull wires are attached to the guide catheter in diametrically opposed locations and another two of the at least four pull wires are attached to the guide catheter in diametrically opposed locations. In other words, the catheter may include two sets of pull wires, each set functioning in an opposing manner as described. When the two sets of pull wires are positioned so that each pull wire is 90 degrees apart, the catheter may be curved so that the distal end is directed from side to side and up and down. In other embodiments, the steering mechanism comprises at least three pull wires, each pull wire symmetrically positioned approximately 120 degrees apart. When tension is applied to any of the pull wires individually, the catheter is curved in the direction of the pull wire under tension. When tension is applied to two pull wires simultaneously, the catheter is curved in a direction between the pull wires under tension. Additional directions may also be achieved by various levels of tension on the pull wires. It may be appreciated that any number, combination and arrangement of pull wires may be used to direct the catheters in any desired direction.

(22) In some embodiments, a portion of one of guide catheter **105** can comprise one or more articulating members. In this case, the at least one pull wire is attached to one of the articulating members so that the curve is formed by at least some of the articulating members. Each pull wire is attached to the catheter at a location chosen to result in a particular desired curvature of the catheter when tension is applied to the pull wire. For example, if a pull wire is attached to the most distal articulating member in the series, applying tension to the pull wire will compress the articulating members proximal to the attachment point along the path of the pull wire. This results in a curvature forming in the direction of the pull wire proximal to the attachment point. It may be appreciated that the pull wires may be attached to any location along the catheter and is not limited

to attachment to articulating members. Typically, the articulating members comprise inter-fitting domed rings but may have any suitable shape.

(23) It may also be appreciated that curves in guide catheter **105** may be formed by any combination of mechanisms. For example, a portion of guide catheter could form a curve by shape memory while a different portion of guide catheter could form a curve by actuation of a steering mechanism.

(24) The steering mechanisms may be actuated by manipulation of actuators located on handle **110**. The handle **110** can be connected with the proximal end of the guide catheter **105** and remains outside of the body. One or more actuators or controls **120** can be provided on handle **110** and may have any suitable form, including buttons, levers, knobs, switches, toggles, dials, or thumbwheels, to name a few. When pull wires are used, each actuator may apply tension to an individual pull wire or to a set of pull wires. The handle may also include one or more locking mechanisms configured to interface with, and selectively lock into place, one or more of the controls **120**.

(25) In at least one embodiment, the handle **110** includes at least one control **120** for actuating and/or adjusting one or more components of a cutting mechanism **130**. As shown in FIG. 1, the cutting mechanism **130** is configured to extend beyond the distal end **115** of the guide catheter **105**. In at least one embodiment, the cutting mechanism **130** is routable through the guide catheter **105** and retractable into the guide catheter **105**. The at least one control **120** may control the cutting mechanism's **130** extension from, and retraction into, the guide catheter **105**. Additionally or alternatively, the at least one control **120** may be configured to provide selective actuation of the cutting mechanism **130**. The at least one control **120** may be operatively connected to one or more additional elements of the cutting mechanism **130**. The cutting mechanism **130** is shown here in generic form as a dashed line, and therefore represents any of the cutting mechanism **130** embodiments described herein.

(26) FIG. 2 is a schematic representation of a human heart with an interventional implant affixed between the anterior and posterior leaflets of the mitral valve. The mitral valve comprises a posterior mitral leaflet **210** and an anterior mitral leaflet **215**. FIG. 2 also shows an interventional implant (e.g., MitraClip®) **220**, which has previously been affixed to the leaflets in an effort to reduce regurgitation by approximating the adjacent leaflets **210** and **215**. As shown, the affixation of implant **220** to the leaflets creates a first orifice **222** and a second orifice **224** located on opposing sides of implant **220** and between the anterior mitral leaflet **215** and the posterior mitral leaflet **210**. And, as discussed above, if further treatment is required in the form of the installation of an artificial, replacement mitral valve, the prior clip implant **220** must be detached from one or both of the leaflets before the replacement valve can be implanted.

(27) In use, the leaflet cutting system **100** can be inserted into, and navigated through a patient's vasculature in a conventional manner to arrive in the patient's heart. As also illustrated in FIG. 2, a distal end portion of the leaflet cutting system **100** can be inserted through the interatrial septum **200** of the heart and positioned above the mitral valve in preparation for a leaflet cutting procedure. In at least one embodiment, the distal end **115** of the guide catheter **105** can include one or more radiopaque or echogenic markers to aid in accurate positioning. One skilled in the art will appreciate that the positioning of the distal end portion of the guide catheter **105** of the leaflet cutting system **100** in FIG. 2 is merely exemplary, and the present disclosure is not limited to the specific positioning shown.

(28) Referring next to FIGS. 3A-3D, which illustrate a first embodiment of a leaflet cutting system **100**. System **100** includes components corresponding to those described above in connection with FIG. 1, which are designated by the same reference numerals throughout. As shown in FIGS. 3A-3D, system **100** also includes a cutting mechanism **130a** positioned within, and routed through, the guide catheter **105**. As shown, the cutting mechanism **130a** may comprise an elongate inner catheter or hypotube **300**, extending from a proximal end (not shown in FIG. 3) to a distal end **302** of the leaflet cutting system **100**. The proximal end of the cutting mechanism **130a** can be

operatively coupled to the handle **110**, and handle **110** provided with controls adapted to manipulate the cutting catheter **130a**, including advancing, retracting and/or rotating the cutting mechanism **130a** relative to the guide catheter **105**.

(29) As shown in FIG. 3A, cutting mechanism **130a** is positioned within the interior of guide catheter **105** during advancement and positioning of the system **100** above the mitral valve **205**. As discussed below, once the distal end **115** of guide catheter **105** is properly positioned above the mitral valve **205** and in proper alignment with interventional implant **220**, the handle **110** can be manipulated to cause cutting mechanism **130a** to advance relative to guide catheter **105** and to pass through the tissue of one of the leaflets **210**, **215**, thereby cutting the affected leaflet and separating the interventional implant **220** from the affected leaflet.

(30) In this embodiment, the cutting mechanism **130a** can have at its distal end **302**, a sharpened end **304** that terminates in a point. The sharpened end **304** may have a circular or oval cross-sectional shape. The sharpened end **304** can also have tapered, sharpened edges or blades adjacent to and extending from the point, which are adapted to slide through the leaflet tissue located between the first and second orifices **222**, **224** and adjacent the interventional implant **220**. In this and other embodiments described herein, the components of cutting mechanism **130a** may be formed from the same or different materials, including but not limited to stainless steel or other metals, Elgiloy®, nitinol, titanium, tantalum, metal alloys or polymers.

(31) FIG. 3B also shows how the hypotube **300** may be advanced to cause the sharpened end **304** to extend from the distal end **115** of the guide catheter **105** and cut a portion of the anterior mitral leaflet **215**. The point of sharpened end **304** helps anchor and stabilize the leaflet tissue as the sharpened, cutting edges of the cutting mechanism **130a** are advanced through and cut the leaflet tissue. The circular or oval cross section of sharpened end **304** of the cutting mechanism **130a** cause the sharpened end to cut through the leaflet tissue in an arc around the interventional implant and extending from the first orifice **222** to the second orifice **224**, as schematically illustrated in FIG. 3D. After the sharpened end **304** of cutting mechanism **130a** advances and cuts through the tissue of the anterior mitral leaflet **215**, the interventional implant **220** may remain attached to the posterior mitral leaflet **210**, as shown in FIG. 3C, thereby reducing the risk that the interventional implant **220** will interfere with functioning of the left ventricular outflow tract. Additionally or alternatively, the posterior mitral leaflet **210** may also be cut in a similar manner with little or acceptable risk of left ventricular outflow tract interference. In at least one embodiment, the interventional implant **220** is removed from the patient.

(32) FIG. 3D is a top perspective view of a mitral valve showing the cut portion of the anterior mitral leaflet **215**. As shown, the sharpened end **304** may cut an arc in the leaflet **210** around the interventional implant **220**. The hypotube **300** may be sized depending on the number of interventional implants **220** that are located within the cardiac valve. Additionally or alternatively, a single hypotube **300** may be used to cut multiple interventional implants **220** within a cardiac valve.

(33) In this and other embodiments described herein, inner catheter or hypotube **300** should preferably have sufficient flexibility as to be able to conform to bends formed by guide catheter **105**. Additional flexibility to accommodate bending may be provided in certain regions of hypotube **300** by a series of laser cuts formed in the outer wall of hypotube **300**. In addition, hypotube **300** should also provide sufficient compressive strength to permit forces to be transmitted through hypotube **300**, from the proximal end to the distal end, sufficient to cause the sharpened end **304** of hypotube **300** to cut through the leaflet tissue.

(34) In an alternate embodiment, an inflator **240** can be attached to the guide catheter **105** at or near its proximal end. The inflator **240** can be configured to create a vacuum transmitted through the lumen of the guide catheter **105**. When the distal end of the guide catheter is positioned against the leaflet, a negative pressure can be applied to the leaflet tissue, thereby stabilizing nearby leaflet tissue located adjacent the interventional implant **220**. While leaflet tissue adjacent to the clip

implant **220** is held in place by the negative vacuum pressure applied through the distal end of the guide catheter, the hypotube **300** can be advanced relative to the guide catheter **105**, thereby causing the sharpened end **304** to pass through and cut the leaflet tissue surrounding the clip implant **220**. In this alternate embodiment, a distal end portion of the guide catheter **105** may comprise various cross-sectional shapes including a U-shape. For example, the distal end **115** of the guide catheter **105** can comprise a flexible U-shape that may collapse when positioned within a delivery sheath and then expand when the guide catheter is advanced to extend beyond the distal end of the delivery sheath.

(35) In another alternate embodiment, in addition to the sharpened edges located at the distal end of the cutting mechanism **130a** used for mechanical cutting, the cutting mechanism **130a** could also include an electrical conductor (not shown) that extends along its entire length, which conductor is electrically coupled at a distal end to the sharpened cutting end **304** and which is also electrically coupled at a proximal end to a source for selectively applying electrosurgical energy, such as an electrosurgical generator. In that case, the cutting of leaflet tissue could be achieved by mechanical cutting, by the application of electrosurgical energy, or a combination of both.

(36) Reference is next made to FIGS. **4A-4F**, which illustrate a second embodiment of a cutting mechanism **130b**. This embodiment also includes components corresponding to those described above in connection with FIG. **1**, which are designated by the same reference numerals throughout. FIG. **4A** is a side sectional view of the cutting mechanism **130b** shown within the distal end **115** of the guide catheter **105**. As shown, the cutting mechanism **130b** may comprise an inner catheter or hypotube **400** having a sharpened, cutting end **402**, which terminates at its distal end in a sharpened point. This embodiment, the sharpened cutting end **402** can include two blades **404** and **406** joined at a pivot point. The two blades **404** and **406** may each comprise a cutting edge disposed on an outer edge of each blade as illustrated in FIG. **4B**. The hypotube **400** may be configured to be selectively extended from the distal end **115** of the guide catheter **105**, thereby allowing the two blades **404** and **406** to engage with a portion of leaflet tissue.

(37) In this embodiment, the blades **404** and **406** can be connected to the hypotube **400** in such a way that, when actuated, the blades pivot outwardly away from one another in an arcuate path in a “reverse-scissor” fashion. In the illustrated embodiment, this can be accomplished as follows. A tip ring **408** can be integrally attached to the distal end of the hypotube. Tip ring **408** can include a center lumen that allows an actuating rod **410** to extend therethrough as shown in FIG. **4B**. The actuating rod **410** can be pivotally connected at its distal end to a center portion of the proximal ends of each of blades **404** and **406** as shown. Links **412** and **414** can also be provided, which connect a distal end of tip ring **408** to an outer portion of the proximal ends of each of blades **404** and **406**. While not explicitly shown in FIG. **4**, hypotube **400** and actuating rod **410** can run the entire length of delivery system **102** and can be operatively coupled at their proximal ends to handle **110**, and handle **110** can be provided with suitable controls to allow manipulation of hypotube **400** and actuating rod **410** relative to one another to cause blades **404** and **406** to open and close relative to one another. It will be appreciated that advancing actuating rod **410** relative to hypotube **400** will cause blades **404** and **406** to open in an arcuate path away from one another, and that withdrawing actuating rod **410** relative to hypotube **400** will cause blades **404** and **406** to close and move toward one another. Alternatively, actuating rod **410** could be held stationary and hypotube **400** moved relative to actuating rod **410** to open and close blades **404** and **406**. Other structures and mechanisms known to those skilled in the art could be adapted and used for opening and closing blades **404** and **406** and are intended to be encompassed within the scope of the invention.

(38) The cross-sectional shape of blades **404** and **406** can be selected to influence that shape of the path they travel as they cut through leaflet tissue. For example, in one embodiment illustrated in FIG. **4C**, blades **404** and **406** can have a relatively straight or flat cross-sectional shape, which will cause the blades to cut through the leaflet tissue in a relatively straight path. Referring to FIG. **4D**,

the cut through the leaflet tissue can be made in multiple segments. For example, with blades **404** and **406** closed, the sharpened end **402** can be inserted at a first insertion point **416a** into and through leaflet **210**. The sharpened point and sharpened edges of sharpened end **402** assist with the ease of insertion and also help stabilize the tissue adjacent to the interventional implant **220**. Once inserted through the tissue, blades **404** and **406** can be actuated by means of handle **110**, acting through hypotube **400** and actuating rod **410**, to cause the blades to open, thereby cutting through the leaflet tissue located to either side of insertion point **416a**. The cutting mechanism can then be withdrawn in a proximal direction, repositioned, re-advanced in a distal direction through the tissue at a second insertion point **418a**, re-actuated, and then at a third insertion point **418c**, and so and so forth until the process completed to create a continuous cut through the affected leaflet tissue around the interventional implant **220** and extending from the first orifice **222** to the second orifice **224**. Alternatively, the opening and closing of blades **404** and **406** can simply be reciprocated as the guide catheter **105** is used to drag the cutting blades through the leaflet tissue.

(39) In an alternate embodiment, as illustrated in FIGS. **4E-4F**, the two blades **404** and **406** can be shaped such that they cut the portion of leaflet tissue in an arcuate path. As best shown in FIG. **4E**, blades **404** and **406** can also be made to have a cross-sectional shape that is circular or oval in shape. As with the above, the pointed end of sharpened end **402** is advanced through an entry point **416a** into the leaflet tissue. However, when actuated, the shape of blades **404** and **406** cause them to travel in an actuate path through the leaflet tissue, as illustrated in FIG. **4F**.

(40) The cut portion in FIG. **4F** is shown on the anterior leaflet **215**, however, the present invention is not limited to the positioning of shape or cut. In at least one embodiment, the posterior mitral leaflet **210** can also be cut. Additionally or alternatively, the interventional implant **220** may be removed from the patient.

(41) Reference is next made to FIGS. **5A-5J**, which illustrate yet another embodiment of the cutting mechanism. In this embodiment, cutting mechanism **130c** may comprise a first and a second delivery catheter **500** and **502**, each comprising a proximal end and a distal end. The first delivery catheter **500** can include at its distal end a pair of grasping arms **504** and **506**, which arms can be rotatably coupled to the distal end of delivery catheter **500** by a rotatable hub **508**. Similarly, second delivery catheter **502** can include at its distal end a pair of grasping arms **510** and **512**, which arms can be rotatably coupled to the distal end of delivery catheter **502** by a rotatable hub **514**. One or more of grasping arms **504**, **506**, **510** and/or **512** may include one or more teeth to prevent slippage when the arms are engaged with leaflet tissue.

(42) A cutting wire **516** may extend from the hub **508** of the first delivery catheter **500** to the **514** of the second delivery catheter **502**. The cutting wire **516** may be configured to selectively provide electrosurgical energy to a secured portion of leaflet tissue, thereby cutting a portion of leaflet tissue when rotated around hubs **508** and **514**. Additionally or alternatively, an adjustment in the tension of cutting wire **516** may cause the wire **516** to engage with and cut the secured leaflet tissue.

(43) As shown in FIG. **5B**, which is a top perspective view of a mitral valve, the first and second delivery catheters **500** and **502** may be configured to extend from the distal end **115** of the guide catheter **105**. The interventional implant **220** creates a first orifice **222** and a second orifice **224** between the anterior mitral leaflet **215** and the posterior mitral leaflet **210** by approximating the adjacent leaflets **210** and **215**. The first delivery catheter **500** may extend toward the first orifice **222** and thereby extend the first grasping arm **504** (not shown in FIG. **4B**) of the first delivery catheter **500** into the first orifice **222** of the mitral valve. Similarly, the second delivery catheter **502** may extend toward the second orifice **224** and thereby extend the first grasping arm **510** (not shown in FIG. **4B**) of the second delivery catheter **502** into the second orifice **224** of the mitral valve. The first grasping arms **504** and **510** may be configured to engage with leaflet tissue on the ventricular side of the mitral valve.

(44) FIGS. **5C** and **5D** are a detailed side views of the distal end of delivery catheter pair **500** and

502 at various stages of positioning and deployment to attach and hold onto a portion of the tissue of one of the leaflets **205**, with one delivery catheter **500** attaching to the leaflet **210** to one side of the interventional implant **220** and the other delivery catheter **502** attaching to the same leaflet **210** on the other side of the interventional implant **220**. Once the guide catheter **105** is properly positioned above and aligned with the interventional implant **220**, delivery catheters **500** and **502** are advanced relative to the guide catheter **105** to extend beyond the distal end **115** of guide catheter **105**. The distal ends of delivery catheters **500** and **502** are advanced until grasping arms **504** and **510** pass through orifices **222** and **224** of the mitral valve on opposite sides of the interventional implant **220**, as shown in FIG. 5C. Guide catheter **105** can then be manipulated to reposition grasping arms **504** and **510** into direct contact with the edges of orifices **222** and **224**, as also shown in FIG. 5C. Once that is done, grasping arms **504** and **506** of delivery catheter **500** can be rotated toward one another to clamp down on the leaflet tissue adjacent the first orifice **222** (and located to one side of the interventional device **220**), and grasping arms **510** and **512** of delivery catheter **502** can similarly be rotated toward one another to clamp down on either side of the leaflet tissue adjacent the second orifice **224** (and located to the other side of the interventional device **220**), as shown in FIG. 5D. Once the leaflet tissue is secured between grasping arms **504** and **506** and between grasping arms **510** and **512**, then cutting wire **516** can be rotated to cut through the leaflet tissue and separate the interventional implant **220** from the affected leaflet, as illustrated in FIGS. 5E and 5F.

(45) FIGS. 5G-5J illustrate one example of a mechanism of actuating the various components of cutting mechanism **130c** described above. However, additional structures and mechanisms known to those skilled in the art could be adapted and used for controlling the selective, relative rotation of grasping arms **504**, **506**, **510** and **512**, as well as that of cutting wire **516**. In one embodiment, each of delivery catheters **500** and **502** can include at their respective distal ends a tip ring **518**, having a plurality of lumens for accepting a plurality of control lines. Each control line, such as control line **522** shown in FIGS. 5H and 5I, can be coupled at the proximal end of the system **100** to a dedicated control located on handle **110**. Separate control lines can be provided to separately and independently control selective movement of each of grasping arms **504**, **506**, **510** and **512**, as well as movement of cutting wire **516**. Certain control lines, such as control line **522**, can extend the entire length of the system and pass around a saddle, such as saddle **520** formed in tip ring **518** between adjacent lumens, as shown in FIGS. 5I and 5J, so that the control line returns in a proximal direction for attachment to one of the rotatable hubs, such as rotatable hub **508**, as shown in FIG. 5J. As further shown in FIG. 5J, control line **522** can be mechanically coupled to the periphery on one side of hub **508**, and a second control line **524** can be mechanically coupled to the periphery of the other side of hub **508**. As will be readily understood, selectively pulling on control line **522** will cause hub **508** to rotate in clockwise direction, and selectively pulling on control line **524** will cause hub **508** to rotate in counter-clockwise direction. By providing multi-part hubs and separate control lines for each component, relative movement and control of grasping arms **500** and **502**, grasping arms **510** and **512**, and cutting wire **516** can be effectuated from the proximal end of system **100** by means of one or more controls **120** provided on handle **110**.

(46) FIGS. 5C-5F illustrate how the grasping arms **504**, **506**, **510** and **512** are configured to secure a portion of the anterior mitral leaflet **215** therebetween and show the cutting wire **516** as cutting a portion of the anterior mitral leaflet **215**. However, in at least one embodiment the posterior mitral leaflet **210** can be cut. The interventional implant **220** may additionally or alternatively be removed. As shown in FIGS. 5E and 5F, the cutting wire **516** may be rotated through the secured portion of the anterior mitral leaflet **215**, thereby cutting the portion of leaflet tissue.

(47) In at least one embodiment, the space between the first and second delivery catheters **500** and **502** may be adjusted based on number of interventional implants **220** within the mitral valve. Additionally or alternatively, the cutting mechanism **130c** may be configured to retract into the distal end **115** of the guide catheter **105** after the portion of leaflet tissue is cut. One skilled in the

art will appreciate that the positioning shown in FIGS. 5C-5F is merely exemplary and the present invention is not limited to the positioning shown.

(48) FIGS. 6A-6F are side sectional views and illustrate yet another alternative exemplary embodiment of a cutting mechanism shown in use in association with a cardiac valve, specifically a mitral valve. In this embodiment, the cutting mechanism **130d** can include a clip grasping structure **602** and a cutting wire **608**. FIG. 6B shows the distal end **115** of the guide catheter **105** positioned above the interventional implant **220**, which approximates the anterior mitral leaflet **215** and posterior mitral leaflet **210**. The cutting wire **608** can include an elongated portion **610** that terminates in a distal lasso or loop portion **612**. The cutting wire **608** may extend from the handle **110** to the distal end **115** of the guide catheter **105**. Further, the cutting wire **608** may be configured to selectively provide electrosurgical energy through the distal loop portion **612** to tissue in contact with distal loop portion **612**. While the cutting wire **608** is shown as a separate component in FIGS. 6A-6D, a cutting electrode could also be integrally formed on the distal end **115** of guide catheter **105** and/or on the interior wall surface of guide catheter **105** adjacent the distal end **115**.

(49) FIGS. 6A-6F also illustrate a clip grasping structure **602** comprising an elongated portion **604** and a distal clamping portion **606**, which may be routed through the guide catheter **105**. As shown in FIG. 6B, the distal clamping portion **606** is extendable distally past the distal end **115** of the guide catheter **105**. The distal clamping portion **606** may extend through the distal loop portion **612** of the cutting wire **608** when extending distally past the distal end **115** of the guide catheter **105**.

(50) The distal clamping portion **606** may be configured to secure an interventional implant **220**, as shown in FIG. 6C. The distal clamping portion **606** may also be retractable proximally into the distal end **115** of the guide catheter **105** with the secured interventional implant **220**. In at least one embodiment, the distal clamping portion **606** extends through the distal loop portion **612** of the cutting wire **608** when retracting proximally into the distal end **115** of the guide catheter **105** with the secured interventional implant **220**. As shown in FIG. 6D, the distal loop portion **612** of the cutting wire **608** may be configured to cut a portion of leaflet tissue when the distal clamping portion **606** of the clip grasping structure **602** retracts proximally into the distal end **115** of the guide catheter **105** with the secured interventional implant **220**. The distal loop portion **612** of the cutting wire **608** may also be selectively constricted to engage with and cut the portion of leaflet tissue. In at least one embodiment, the cutting wire **608** is configured to detach the interventional implant **220** from surrounding leaflet tissue.

(51) FIGS. 6E and 6F illustrate one example of a mechanism of actuating the distal clamping portion **606** of grasping structure **602**. However, additional structures and mechanisms known to those skilled in the art could be adapted and used for selectively opening, closing and clamping distal clamping portion **606**. In this embodiment, grasping mechanism **602** can include an elongate hypotube **614**, an actuating rod **616**, a pair of grasping arms **618** and **620**, and a pair of links **622** and **624**. A proximal end of each grasping arm **618** and **620** can be pivotally coupled to the distal end of actuating rod **616**. Links **622** and **624** can each be pivotally connected at one end to the distal end of hypotube **614** and can be pivotally connected at the other end to an intermediate location on grasping arms **618** and **620**, respectively. It should be further understood that hypotube **614** and actuating rod **616** can extend the entire length of the system and can be operatively coupled at their proximal ends to suitable controls **120** located on handle **110** to control movement of hypotube **614** and actuating rod **616** relative to one another and, thus, causing grasping arms to open and close. The tissue engaging surface of grasping arms **618** and **620** can also include one or more teeth to engage leaflet tissue adjacent the interventional implant and prevent slippage.

(52) Reference is next made to FIGS. 7A-7D, which illustrate still yet another embodiment of a cutting mechanism for use in detaching an interventional device from one or more leaflets of a mitral valve. Cutting mechanism **130e** can include an inner catheter or hypotube **700**, a clip grasping structure **702** and an elongated cutter **704**. As shown in Figures 7A-7C, the distal end **115** of the guide catheter **105** may be positioned above the interventional implant **220**. The elongated

cutter **704** can be configured to be positioned within the lumen formed between the outer wall of the hypotube **700** and the inner wall of the guide catheter **105** and to selectively extend from the distal end **115** of the guide catheter **105**. Cutter **704** can terminate at its distal end in a sharpened point adapted to pierce through leaflet tissue. Cutter **704** can also include one or more sharpened edges **714** (adapted to mechanically cut through the leaflet tissue) that extend along one or both sides of a distal section thereof. The elongated cutter **704** may rotate around the clip grasping structure in an arcuate path, thereby slicing through a portion of leaflet tissue. In at least one embodiment the elongated cutter **704** can be composed of a heat shaped material.

(53) FIGS. 7A-7C further illustrate that hypotube **700** extends through the guide catheter **105** and can be positioned at the distal end **115** of the guide catheter **105**. Clip grasping structure **702** may be routable through the hypotube **700** and positioned at a distal end of the hypotube **700**. In at least one embodiment, the clip grasping structure **702** can comprise two flexible prongs **706** and **708**, each of which preferably includes a sharpened tip adapted to pierce through the tissue of the mitral valve leaflets. The clip grasping structure **702** may comprise an elongated portion **710** and a distal clamping portion **712**. As shown in FIGS. 7A and 7B, the distal clamping portion **712** of the clip grasping structure **702** may be extendable distally past the distal end **115** of the guide catheter **105** and the distal end of the hypotube **700**. In at least one embodiment, the distal clamping portion **712** can be made of a semi-rigid, resilient metal. Once advanced beyond the distal end and freed from the constraints of hypotube **700**, the distal clamping portion **712** can expand laterally as illustrated in FIG. 7A. Once in this configuration, the distal clamping portion **712** of the clip grasping structure **702** may be positioned around the interventional implant **220**, and the assembly may then be advanced distally, causing the prongs **706** and **708** of distal clamping portion **712** to puncture through leaflet tissue located to either side of the interventional device **220**. Then, as shown in FIG. 7B, hypotube can be advanced distally from the distal end **115** of the guide catheter **105** relative to the clip grasping structure **702**, such that the hypotube **700** encloses a proximal portion of the distal clamping portion **712** of the clip grasping structure **702** within the hypotube **700**, which also draws the prongs **706** and **708** closer together, thereby causing the distal clamping portion **712** of the clip grasping structure **702** to secure an interventional implant **220**. The elongated cutter **704** may then be advanced relative to the guide catheter **105**, hypotube **700** and clip grasping structure **702** to extend beyond the distal end **115** of the guide catheter **105** and engage with and cut the leaflet tissue. More specifically, the distal end of cutter **704** may be advanced through the first orifice **222** or the second orifice **224**, and then the system can be rotated about its central axis, causing cutter **704** to travel in an arcuate path around the interventional device **220**, slicing through the leaflet tissue located adjacent to the interventional device **220** as it travels from one orifice to the other. Of course, the cutter **704** could also be advanced through the leaflet tissue and then the system could be rotated back and forth until the entire cut from one orifice to the other orifice is complete. In addition, with this embodiment it is possible to cut through the tissue of both leaves of the mitral valve, thereby completely separating the interventional device **220** from the mitral valve. In that case, as shown in FIG. 7C, the hypotube **700** and distal clamping portion **712** of the clip grasping structure **702** can be retracted in a proximal direction relative to the guide catheter and into the distal end **115** of the guide catheter **105**, thereby securing interventional implant **220**. FIG. 7D shows the cut portion **610** of the anterior mitral leaflet **215**, however the present invention is not limited to the positioning or shape of cut.

(54) Finally, reference is next made to FIGS. 8A-8C, which illustrate still yet another embodiment of a cutting mechanism for use in detaching an interventional device from one or more leaflets of a mitral valve. In this embodiment, cutting mechanism **130f** can include an inner catheter or hypotube **800** and a flexible stabilizing rod or dilator **802**. As shown, stabilizing rod **802** extends through the inner lumen of hypotube **800** and can be selectively advanced and/or retracted relative to hypotube **800**. The guide catheter is not shown in FIGS. 8A and 8C, but it should be understood that cutting mechanism **130f** is intended for use within a suitable guide catheter to position the

distal end of cutting mechanism **130f** above the interventional device **220**. Once properly positioned, stabilizing rod **802** can be advanced relative to the guide catheter and hypotube **800** so as to extend beyond the distal end of hypotube **800** by a predetermined distance. With the stabilizing rod **802** advanced, the entire system can be advanced distally, until the distal end of stabilizing rod **802** engages the fold of leaflet tissue located opposite the interventional clip **220**, which fold is created when the posterior and anterior leaflets **210** and **215** are brought and held together by interventional device **220**.

(55) As shown in FIGS. **8A** and **8C**, hypotube **800** can include one or more lumens formed in the wall of hypotube **800**, such as lumen **804** shown in FIG. **8A** and lumens **804** and **806** shown in FIG. **8B**. As illustrated, each lumen **804** and **806**, extends from the proximal end of hypotube **800** and then exits at an angle laterally outwardly from the side wall of hypotube **800** near the distal end of hypotube **800**. Positioned within each lumen **804** and **806** is an elongated cutter **808** and **810**. Each cutter **808** and **810** preferably terminates in a cutting blade portion **812** and **184** at its distal end, and can be selectively advanced and retracted relative to hypotube **800**. As shown in FIGS. **8A** and **8C**, when cutters **808** and **810** are advanced to extend beyond hypotube **800**, the distal ends of cutter **808** and **810** splay radially outwardly at an angle relative to the central axis of hypotube **800**.

(56) In the case of the embodiment shown in FIG. **8A**, once the distal end of the stabilizing rod **802** is firmly engaged against the leaflet fold opposite the interventional device **220** as described above, then cutter **808** can be advanced to extend out and beyond the end of hypotube **800**. In some instances, the advancement of cutter **808** will also extend through one of the orifices **210** or **215**, but it may also be necessary to further advance hypotube **800** relative to stabilizing rod **802** until cutter **808** extends into orifice **222** or **224**. At that point, hypotube **800** alone or hypotube **800** and the guide catheter **105** together can be rotated around stabilizing rod **802**, thereby causing the cutting blade **812** of cutter **808** to slice through the leaflet tissue in an arcuate path around interventional device **220** as schematically illustrated in FIG. **8B**. With a single cutter **808**, a rotation of 180 degrees should be sufficient to detach one leaflet from the interventional device **220**, and a rotation of 360 degrees should be sufficient to detach both leaflets from the interventional device. As will be appreciated, the embodiment shown in FIG. **8C** operates in a substantially similar manner, except that each cutter **808** and **810** can simultaneously be inserted through both orifices, and cutting of both leaflets can be completed with a single rotation of 180 degrees.

(57) In describing the various embodiments above, the description may at times have explicitly discussed one particular mitral valve leaflet, such as anterior leaflet **215**. It should be understood and appreciated, however, that the invention is not intended to be limited to either specific leaflet, but instead can be used to cut either anterior leaflet **215**, posterior leaflet **210**, or both.

(58) It should also be understood that the order of manipulation of components of the various embodiments as described above are provided as representative examples only, and changes in the order of manipulation that may be readily understood by those skilled in the art are intended to be encompassed within the scope of this disclosure.

(59) Further still, in addition to the embodiments described above, it should also be understood that individual components from one embodiment could also be combined with and/or substituted for a comparable component described in a different embodiment. For example, the stabilizing rod **802** disclosed in relation to the embodiment shown in FIGS. **8A-8C** could also be incorporated into one or more of the other embodiments, such as, for example, the embodiment illustrated in FIGS. **3A-3D**. Similarly, the cutting blades and structure shown in FIGS. **8A** and **8C** could be substituted in place of the cutting blade shown in FIGS. **7A-7C**, or vice versa.

(60) Similarly, while many of the embodiments discussed above contemplate mechanical cutting of leaflet tissue by means of sharpened edges of a cutting element, it should be further understood that such embodiments could also be adapted to include suitable electrical connections between the cutting element and a source of electrosurgical energy so that such cutting elements may

accomplish cutting of tissue by mechanical cutting, by the application of electrosurgical energy to surrounding tissue through the cutting element, or by a combination of both.

(61) Also, with any or all of the foregoing embodiments, one or more components of the leaflet cutting system can also include one or more radiopaque and/or echogenic markers to aid in the visualization of such components during a procedure. For example, one or more radiopaque and/or echogenic markers can be provided on the distal end **115** and/or the steerable portion **117** of the guide catheter **105**. Similarly, one or more radiopaque and/or echogenic markers can also be provided on various components of the different embodiments of the cutting mechanisms described above, including, but not limited to such markings being provided on the distal ends of the inner catheter, hypotube, clip grasping structures, cutting blades, stabilizing rods, etc.

(62) One skilled in the art will appreciate that the present invention is not limited to use within the mitral valve. The cardiac valve could also be the tricuspid aortic, pulmonic valve, etc. More generally, the embodiments described herein may be applied in other implementations involving removal of a previously implanted or deployed device from tissue. Further, although figures show the guide catheter **105** extending through the interatrial septum **200**, the present invention is not limited to use via a transseptal approach. Any suitable delivery approach may be used, including transfemoral, radial, transjugular, or transapical.

(63) Following are some further example embodiments of the invention. These are presented only by way of example and are not intended to limit the scope of the invention in any way.

(64) Embodiment 1. A system for cutting leaflet tissue at a cardiac valve, comprising a guide catheter having a proximal end and a distal end, wherein the distal end of the guide catheter is steerable to a position above a cardiac valve, a handle coupled to the proximal end of the guide catheter, the handle comprising at least one control configured to steer the guide catheter to the position above the cardiac valve, and a cutting mechanism routable through the guide catheter and able to be positioned at the distal end of the guide catheter, the cutting mechanism configured to cut a portion of leaflet tissue of the cardiac valve.

(65) Embodiment 2. The system of embodiment 1, wherein the at least one control is further configured to provide selective actuation of the cutting mechanism.

(66) Embodiment 3. The system in any of embodiments 1 to 2, wherein the guide catheter is introduced transseptally.

(67) Embodiment 4. The system in any of embodiments 1 to 3, further comprising an inflator attached to the guide catheter, the inflator configured to hold the leaflet in place by applying negative pressure.

(68) Embodiment 5. The system in any of embodiments 1 to 4, wherein the guide catheter is U-shaped.

(69) Embodiment 6. The system in any of embodiments 1 to 5, wherein: the cutting mechanism comprises two blades joined at a pivot point, wherein the two blades each comprise a cutting edge disposed on an outside surface of each of the two blades and the two blades are oriented and shaped such that they cut the portion of leaflet tissue of the cardiac valve in a predefined arc; and the cutting mechanism is configured to extend from the distal end of the guide catheter.

(70) Embodiment 7. The system in any of embodiments 1 to 6, wherein: the cutting mechanism comprises a hypotube with a sharpened end that terminates in a point, wherein the sharpened end is configured to cut the portion of leaflet tissue of the cardiac valve in a predefined arc; and the cutting mechanism is configured to extend from the distal end of the guide catheter.

(71) Embodiment 8. The system in any of embodiments 1 to 7, wherein: the cutting mechanism comprises a first and a second delivery catheter, each comprising a proximal end and a distal end; each of the first and second delivery catheters comprises a rotatable paddle joined to the delivery catheter by a joint at the distal end of the delivery catheter; each of the first and second delivery catheters comprises a gripping mechanism rotatable around the joint and positioned between the first delivery catheter and the rotatable paddle of the first delivery catheter, and the second delivery

catheter and the rotatable paddle of the second delivery catheter; and a wire extending from the joint of the first delivery catheter to the joint of the second delivery catheter.

(72) Embodiment 9. The system in any of embodiments 1 to 8, wherein: the first delivery catheter is configured to extend from the distal end of the guide catheter thereby extending the rotatable paddle of the first delivery catheter into a first orifice of the cardiac valve, the gripping mechanism of the first delivery catheter and the rotatable paddle of the first delivery catheter are configured to secure leaflet tissue therebetween, the second delivery catheter is configured to extend from the distal end of the guide catheter thereby extending the rotatable paddle of the second delivery catheter into a second orifice of the cardiac valve, and the gripping mechanism of the second delivery catheter and the rotatable paddle of the second delivery catheter are configured to secure leaflet tissue therebetween.

(73) Embodiment 10. The system in any of embodiments 1 to 9, wherein the gripping mechanism of each of the first and second guide catheters secures the leaflet tissue on an atrial side, and the rotatable paddle of each of the first and second guide catheters secures the leaflet tissue on a ventricular side.

(74) Embodiment 11. The system in any of embodiments 1 to 10, wherein: the wire is configured to rotate around a hub within the joint of each of the first and second delivery catheters; and the wire is configured to selectively provide radio frequency current energy to the portion of leaflet tissue of the cardiac valve, thereby cutting the portion of leaflet tissue when rotated around the hub within the joint of each of the first and second delivery catheters.

(75) Embodiment 12. The system in any of embodiments 1 to 11, further comprising a clip grasping structure comprising an elongated portion and a distal clamping portion; wherein: the distal clamping portion of the clip grasping structure is extendable distally past the distal end of the guide catheter and configured to secure an interventional implant that approximates adjacent leaflets of the cardiac valve; and the distal clamping portion of the clip grasping structure is retractable proximally into the distal end of the guide catheter with the secured interventional implant.

(76) Embodiment 13. The system in any of embodiments 1 to 12, wherein the distal clamping portion of the clip grasping structure comprises two prongs.

(77) Embodiment 14. The system in any of embodiments 1 to 13, wherein: the cutting mechanism comprises a cutting wire comprising an elongated portion that terminates in a distal loop portion, the cutting wire extends from the handle to the distal end of the guide catheter, and the distal clamping portion of the clip grasping structure extends through the distal loop portion of the cutting wire when extending distally past the distal end of the guide catheter and retracting proximally into the distal end of the guide catheter with the secured interventional implant.

(78) Embodiment 15. The system in any of embodiments 1 to 14, wherein: the cutting wire is configured to selectively provide radio frequency current energy, and the distal loop portion of the cutting wire is configured to cut the portion of leaflet tissue when the distal clamping portion of the clip grasping structure retracts proximally into the distal end of the guide catheter with the secured interventional implant.

(79) Embodiment 16. The system in any of embodiments 1 to 15, wherein the cutting wire is configured to detach the interventional implant from the surrounding leaflet tissue.

(80) Embodiment 17. The system in any of embodiments 1 to 16, further comprising a hypotube routable through the guide catheter and able to be positioned at the distal end of the guide catheter, wherein the clip grasping structure is further routable through the hypotube and able to be positioned at a distal end of the hypotube.

(81) Embodiment 18. The system in any of embodiments 1 to 17, wherein the hypotube is configured to extend from the distal end of the guide catheter when the distal clamping portion of the clip grasping structure is extended distally past the distal end of the guide catheter such that the hypotube encloses a proximal portion of the distal clamping portion of the clip grasping structure

within the hypotube, thereby causing the distal clamping portion of the clip grasping structure to secure an interventional implant.

(82) Embodiment 19. The system in any of embodiments 1 to 18, wherein the cutting mechanism comprises an elongated cutter configured to extend from the distal end of the guide catheter and rotate around a horizontal arc, thereby cutting the portion of leaflet tissue.

(83) Embodiment 20. The system in any of embodiments 1 to 19, wherein the elongated cutter is configured to selectively provide radio frequency current energy to the portion of leaflet tissue of the cardiac valve, thereby cutting the portion of leaflet tissue when rotated around the horizontal arc.

(84) Embodiment 21. A method of cutting leaflet tissue at a cardiac valve within a body, comprising positioning a guide catheter, having a proximal and a distal end such that the distal end of the guide catheter is positioned at a cardiac valve, routing a cutting mechanism through the guide catheter such that the cutting mechanism extends to the distal end of the guide catheter, wherein the cardiac valve is associated with an interventional implant that approximates adjacent leaflets of the cardiac valve, and a cutting mechanism extends from the guide catheter; and actuating the cutting mechanism to cut at a portion of least one leaflet of the approximated adjacent leaflet.

(85) The present invention may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes that come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A system for severing leaflet tissue at a cardiac valve, comprising: a guide catheter having a proximal end and a distal end, wherein the distal end of the guide catheter is steerable to a position above a cardiac valve; a handle coupled to the proximal end of the guide catheter, the handle comprising at least one control configured to steer the guide catheter to the position above the cardiac valve; a cutting mechanism routable through the guide catheter and able to be positioned at the distal end of the guide catheter, the cutting mechanism configured to sever a portion of leaflet tissue of the cardiac valve, wherein the cutting mechanism comprises: a hypotube having a proximal end and a distal end with a lumen extending therebetween; a tapering blade portion extending distally from a portion of the circumference of the distal end of the hypotube, the blade portion terminating at a distal most point longitudinally aligned with a first edge along the circumference of the hypotube; wherein arcuate sharpened outer edges extend from the distal most point, the sharpened edges tapering proximally and radially outward along a circumference of the hypotube, the sharpened edges terminating at a second edge diametrically opposed from the first edge; and wherein the cutting mechanism is configured to sever the portion of leaflet tissue of the cardiac valve in a predefined arc around an interventional implant.

2. The system of claim 1, wherein the at least one control is further configured to provide selective actuation of the cutting mechanism.

3. The system of claim 1, wherein the guide catheter is introduced transseptally.

4. The system of claim 1, further comprising an inflator attached to the guide catheter, the inflator configured to hold the leaflet in place by applying negative pressure.

5. The system of claim 1, wherein the guide catheter is U-shaped.

6. The system of claim 1, wherein: the cutting mechanism comprises two blades joined at a pivot point, wherein the two blades each comprise a cutting edge disposed on an outside surface of each of the two blades and the two blades are oriented and shaped such that they sever the portion of leaflet tissue of the cardiac valve in a predefined arc; and the cutting mechanism is configured to

extend from the distal end of the guide catheter.

7. The system of claim 1, wherein the cutting mechanism is configured to extend from the distal end of the guide.

8. The system of claim 1, wherein: the cutting mechanism comprises a first and a second delivery catheter, each comprising a proximal end and a distal end; each of the first and second delivery catheters comprises a rotatable paddle joined to the delivery catheter by a joint at the distal end of the delivery catheter; each of the first and second delivery catheters comprises a gripping mechanism rotatable around the joint and positioned between the first delivery catheter and the rotatable paddle of the first delivery catheter, and the second delivery catheter and the rotatable paddle of the second delivery catheter; and a wire extending from the joint of the first delivery catheter to the joint of the second delivery catheter.

9. The system of claim 8, wherein: the first delivery catheter is configured to extend from the distal end of the guide catheter thereby extending the rotatable paddle of the first delivery catheter into a first orifice of the cardiac valve; the gripping mechanism of the first delivery catheter and the rotatable paddle of the first delivery catheter are configured to secure leaflet tissue therebetween; the second delivery catheter is configured to extend from the distal end of the guide catheter thereby extending the rotatable paddle of the second delivery catheter into a second orifice of the cardiac valve; and the gripping mechanism of the second delivery catheter and the rotatable paddle of the second delivery catheter are configured to secure leaflet tissue therebetween.

10. The system of claim 9, wherein the gripping mechanism of each of the first and second guide catheters secures the leaflet tissue on an atrial side, and the rotatable paddle of each of the first and second guide catheters secures the leaflet tissue on a ventricular side.

11. The system of claim 9, wherein: the wire is configured to rotate around a hub within the joint of each of the first and second delivery catheters; and the wire configured to selectively provide radio frequency current energy to the portion of leaflet tissue of the cardiac valve, thereby cutting the portion of leaflet tissue when rotated around the hub within the joint of each of the first and second delivery catheters.

12. The system of claim 1, further comprising a clip grasping structure comprising an elongated portion and a distal clamping portion; wherein: the distal clamping portion of the clip grasping structure is extendable distally past the distal end of the guide catheter and configured to secure an interventional implant that approximates adjacent leaflets of the cardiac valve; and the distal clamping portion of the clip grasping structure is retractable proximally into the distal end of the guide catheter with the secured interventional implant.

13. The system of claim 12, wherein the distal clamping portion of the clip grasping structure comprises two prongs.

14. The system of claim 12, wherein: the cutting mechanism comprises a cutting wire comprising an elongated portion that terminates in a distal loop portion; the cutting wire extends from the handle to the distal end of the guide catheter; and the distal clamping portion of the clip grasping structure extends through the distal loop portion of the cutting wire when extending distally past the distal end of the guide catheter and retracting proximally into the distal end of the guide catheter with the secured interventional implant.

15. The system of claim 14, wherein: the cutting wire is configured to selectively provide radio frequency current energy; and the distal loop portion of the cutting wire is configured to sever the portion of leaflet tissue when the distal clamping portion of the clip grasping structure retracts proximally into the distal end of the guide catheter with the secured interventional implant.

16. The system of claim 14, wherein the cutting wire is configured to detach the interventional implant from surrounding leaflet tissue.

17. The system of claim 12, further comprising a hypotube routable through the guide catheter and able to be positioned at the distal end of the guide catheter, wherein the clip grasping structure is further routable through the hypotube and able to be positioned at a distal end of the hypotube.

18. The system of claim 17, wherein the hypotube is configured to extend from the distal end of the guide catheter when the distal clamping portion of the clip grasping structure is extended distally past the distal end of the guide catheter such that the hypotube encloses a proximal portion of the distal clamping portion of the clip grasping structure within the hypotube, thereby causing the distal clamping portion of the clip grasping structure to secure an interventional implant.
19. The system of claim 17, wherein the cutting mechanism comprises an elongated cutter configured to extend from the distal end of the guide catheter and rotate around a horizontal arc, thereby cutting the portion of leaflet tissue.
20. The system of claim 19, wherein the elongated cutter is configured to selectively provide radio frequency current energy to the portion of leaflet tissue of the cardiac valve, thereby cutting the portion of leaflet tissue when rotated the horizontal arc.
21. A method of cutting leaflet tissue at a cardiac valve within a body, comprising: positioning a guide catheter, having a proximal and a distal end such that the distal end of the guide catheter is positioned at a cardiac valve; routing a cutting mechanism through the guide catheter such that the cutting mechanism extends to the distal end of the guide catheter, wherein the cardiac valve is associated with an interventional implant that approximates adjacent leaflets of the cardiac valve, and a cutting mechanism extends from the guide catheter, and the cutting mechanism comprises: a hypotube having a proximal end and a distal end with a lumen extending therebetween; a tapering blade portion extending distally from a portion of the circumference of the distal end of the hypotube, the blade portion terminating at a distal most point longitudinally aligned with a first edge along the circumference of the distal end of the hypotube; wherein arcuate sharpened outer edges extend from the distal most point, the sharpened outer edges tapering proximally and radially outward along a circumference of the hypotube, terminating at a second edge diametrically opposed from the first edge, wherein the cutting mechanism is configured to sever the portion of leaflet tissue of the cardiac valve in a predefined arc; and actuating the cutting mechanism to sever at a portion of at least one leaflet of the approximated adjacent leaflet.
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